

FINOPS - “Automated Intelligent Document Processing system for Financial operation(Amount Payable)”

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INTRODUCTION

- ▶ In today's digital world, many companies still use traditional methods in their Accounts Payable (AP) process. This system is heavily dependent on manual work, where employees enter invoice data by hand. This creates several serious problems:
- ▶ It is highly reliant on manual data entry, making the process slow and tiring.
- ▶ It is prone to human errors, such as typing mistakes and incorrect calculations.
- ▶ It is vulnerable to duplicate payment fraud, where the same invoice may be paid more than once.
- ▶ It involves time-consuming dispute resolution, requiring manual communication with vendors.
- ▶ Because of these issues, companies face financial losses, reduced efficiency, and unnecessary workload. Therefore, there is a strong need for an intelligent and automated system to improve accuracy, speed, and security

The Problem (Existing System)

- ▶ The current Accounts Payable system faces serious challenges due to its dependence on manual processes.
- ▶ Employees spend a significant amount of time entering invoice details manually, which slows down operations and increases fatigue. As a result, human errors such as incorrect data entry and miscalculations are very common. Additionally, the system lacks proper mechanisms to detect duplicate invoices, making it vulnerable to fraud and repeated payments. Resolving disputes with vendors also takes a long time, as it requires manual checking and communication. Overall, this outdated system reduces efficiency and increases financial risk for organizations.

The Proposed Solution

- ▶ To overcome these challenges, the proposed system introduces a smart and automated solution for handling financial operations.
- ▶ It provides a simple and user-friendly interface where users can easily upload invoices without any training. The system uses MD5 hashing to identify and block duplicate invoices instantly. With the help of advanced AI technology, it can automatically read and extract data from invoices, even if they are unstructured. Additionally, it includes an automated auditing system that verifies calculations and ensures compliance with company policies, reducing errors and improving accuracy.

Technology Stack

- ▶ The system is built using modern and efficient technologies. The frontend is developed using Streamlit, which provides a simple and interactive user interface. The backend is powered by Python, which handles all the business logic and processing.
- ▶ For AI-based data extraction, the system uses Google Gemini API. The data is stored in a MySQL database, which is properly structured for efficiency and reliability. Additional libraries such as hashlib, smtplib, and pandas are used to handle hashing, email communication, and data processing tasks.

Limitations & Future Scope

- ▶ Although the system offers many advantages, it also has some limitations. It requires an internet connection to use cloud-based AI services, and its performance depends on the quality of uploaded files. In the future, the system can be enhanced by integrating it with banking and payment gateways to enable direct payments. Machine learning models can also be added for financial forecasting and budgeting. Additionally, a mobile application can be developed to allow users to scan and upload receipts easily from their smartphones

Conclusion

- In conclusion, the proposed system provides an efficient and intelligent solution to modernize financial operations. It eliminates the need for manual data entry, reduces errors, and prevents duplicate payment fraud. By automating repetitive tasks, it allows employees to focus on more important and strategic work. The system not only improves accuracy and speed but also enhances overall productivity and security in financial processes.

THANK YOU